

NUMBER THEORY



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MODULE 2

The Integers

SUBMODULE 1

Integer Representations and Operations



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OBJECTIVES

- Explain how integers are represented using base expansions.
- Recall number systems conversion.
- Explain how signed and unsigned integers are represented by computers.



Integer Representations and Operations

What is integer representation about?

- It is all about how integers are handled by the computer.
 - How are positive and negative integers represented?
 - How are floating point numbers represented?
 - How are addition, subtraction, multiplication and division being processed?



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Integer Representations and Operations

- In daily life, decimal notation is used to represent integers.
- Numbers are written out using digits to represent powers of ten.

Example: 37465 means $(3 \times 10^4) + (7 \times 10^3) + (4 \times 10^2) + (6 \times 10) + 5$

- Decimal notation is an example of a *positional number system* using ten as the base, in which the position a digit occupies in a representation determines the quantity it represents
- Any positive integer can be used as a base.
- Computers use base 2, base 8 and base 16 as representations of integers.



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Integer Representations and Operations

Any integer n can be uniquely expressed in any base b by the following expression.

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_2 b^2 + a_1 b + a_0$$

In the expression, each coefficient a_i is an integer between 0 and $b - 1$ inclusive.

Note: b – base or radix of the expansion

Base 2 expansions → Binary expansions

Base 8 expansions → Octal expansions

Base 16 expansions → Hexadecimal or Hex expansions

Examples: $(236)_7 = 2 \cdot 7^2 + 3 \cdot 7 + 6$

$$(10010011)_2 = 1 \cdot 2^7 + 1 \cdot 2^4 + 1 \cdot 2^1 + 1$$



Integer Representations and Operations

- Computers represent numbers internally by using a series of “switches” that may be either “on” (represented by the digit 1) or “off” (represented by the digit 0), which is why computers use binary expansions to represent integers internally.
- Computers use base 8 or base 16 for display purposes.
- Base 8 notation has 8 digits denoted by 0, 1, 2, 3, 4, 5, 6 and 7.
- Base 16 notation has 16 digits denoted by numbers 0 to 9 and letters A to F to represent numbers 10 to 15 (written in decimal notation).

Integer Representations and Operations

Base b Expansion of an Integer n (Base 10 to base b)

- Divide n by b to obtain a quotient q_0 and remainder a_0 , that is

$$n = bq_0 + a_0, \text{ where } 0 \leq a_0 < b.$$

- The remainder a_0 is the rightmost digit in the base b expansion of n .
- Next, divide q_0 by b to obtain

$$q_0 = bq_1 + a_1, \text{ where } 0 \leq a_1 < b.$$

- The remainder a_1 is the second digit from the right in the base b expansion of n .
- Continue the process until quotient becomes zero.
- Write the remainders upwards.



Integer Representations and Operations

Base b Expansion of an Integer n (Base 10 to base b)

Example: Find the base 2 expansion of 1864.

$$\begin{aligned}1864 &= 2 \cdot 932 + 0, \\932 &= 2 \cdot 466 + 0, \\466 &= 2 \cdot 233 + 0, \\233 &= 2 \cdot 116 + 1, \\116 &= 2 \cdot 58 + 0, \\58 &= 2 \cdot 29 + 0, \\29 &= 2 \cdot 14 + 1, \\14 &= 2 \cdot 7 + 0, \\7 &= 2 \cdot 3 + 1, \\3 &= 2 \cdot 1 + 1, \\1 &= 2 \cdot 0 + 1.\end{aligned}$$



$$(1864)_{10} = (11101001000)_2$$



Integer Representations and Operations

Conversion from base b to base 10

- Multiply each binary digit to b raised to the position of the base digit. The rightmost base digit has a position value of 0.
- Add all the products to get the resulting decimal number.

Example:

$$\begin{aligned}(A35B0F)_{16} &= 10 \cdot 16^5 + 3 \cdot 16^4 + 5 \cdot 16^3 + 11 \cdot 16^2 + 0 \cdot 16 + 15 \\ &= (10705679)_{10}.\end{aligned}$$

Integer Representations and Operations

Conversion from base 16 to base 2

- Write each hex digit as a block of four binary digits according to the correspondences given in the following table.

Hex Digit	Binary Digits	Hex Digit	Binary Digits
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

Example:

$$(2FB3)_{16} = (10\textcolor{blue}{1111}\textcolor{green}{1011}\textcolor{red}{0011})_2$$



Integer Representations and Operations

Conversion from base 2 to base 16

- Break the binary digits into blocks of four, starting from the right. Add leading zeroes if necessary.
- Translate each block according to the correspondences given in the following table:

Hex Digit	Binary Digits	Hex Digit	Binary Digits
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

Example:

$$\begin{array}{cccc} & & & (11110111101001)_2 \\ & & \downarrow & \\ 0011 & 1101 & 1110 & 1001 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & D & E & 9 \\ = (3DE9)_{16} \end{array}$$



Integer Representations and Operations

Conversion from base 8 to base 2

- Write each octal digit as a block of three binary digits according to the correspondences given in the following table.

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Example:

$$(234)_8 = (10011100)_2$$



Integer Representations and Operations

Conversion from base 2 to base 8

- Break the binary digits into blocks of three, starting from the right. Add leading zeroes if necessary.
- Translate each block according to the correspondences given in the following table:

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Example:

$$\begin{array}{ccc} & (1101110)_2 & \\ & \downarrow & \\ 001 & 101 & 110 \\ \downarrow & \downarrow & \downarrow \\ 1 & 5 & 6 \\ = (156)_8 & & \end{array}$$



Integer Representations and Operations

Activity 1

1. Convert $(1999)_{10}$ to base 7 notation.
2. Convert $(10101111)_2$ to base 10 notation.
3. Convert $(9A0B)_{16}$ to base 2 notation.
4. Convert $(743)_8$ to base 2 notation.



Integer Representations and Operations

How does the computer represent integers?

- Computers represent integers as binary numbers.
- A binary number can only be represented by a 0 or 1, so a minus sign (-) cannot be used to denote a negative number.
- Therefore, no specialized symbol can be used so it must be either a 0 or 1.
- Most Significant Bit (MSB) or Most Significant Digit (MSD) is the first digit (from left) in a binary number. More accurately the digit with the highest radix (base) value.
- Least Significant Bit (LSB) or Least Significant Digit (LSD) is the digit of a binary number with the least radix value, probably the last digit.
- In a binary number, one digit is called bit.
- Computer memory is a place that store data and it has a fixed length of bits.
 - Byte = 8 bits = 8 numbers in binary sequence
 - Word = 16 bits = 16 numbers in binary sequence
 - Double Word = 32 bits = 32 numbers in binary sequence



Integer Representations and Operations

Types of Integers

1. Unsigned Integers

- Represent only zero and positive numbers.
- The value of an unsigned integer interpreted as the “magnitude of underlying binary pattern or the normal way binary numbers are converted to decimal.

Example: 8bit binary number 0100 0010B (B = Binary)
 $0100\ 0010B = 2 \times 2^1 + 2 \times 2^6 = 4 + 64 = 68D$ (D = Decimal)

8bit = 0000 0000

16bit = 0000 0000 0000 0000

32bit = 0000 0000 0000 0000 0000 0000 0000 0000

64bit = 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

All above numbers represent zero in different bit systems. The bit length of each system is different.



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Integer Representations and Operations

n-bit Unsigned Integers

- An n-bit pattern can represent 2^n distinct integers. This range from 0 to $(2^n)-1$

n-bit	Minimum	$2^n - 1$	Maximum
8bit	0	$2^8 - 1$	255
16bit	0	$2^{16} - 1$	65,535
32bit	0	$2^{32} - 1$	4,294,967,295
64bit	0	$2^{64} - 1$	18,446,744,073,709,551,615

When you are programming you need to select the correct and appropriate bit size to store integers. If you want to store small numbers less than 255, you can select 8bit number, selecting 32bit length to store small numbers is a waste of resources.



Integer Representations and Operations

Types of Integers

2. Signed Integers

- Represent only zero, positive and negative numbers.
- The MSB represent the sign if the integer. Zero (**0**) to represent positive (+) sign and One (**1**) to represent negative (-) sign.
- The magnitude of the integer represent differently in different schemas.
- There are three schemas to represent signed integers:
 - Sign and Modulus or Sign Magnitude Representation
 - One's complement Representation
 - Two's complement Representation (The method used by computers.)

Example of Signed Integers:

+2344 is a positive integer

-23434 is a negative integer



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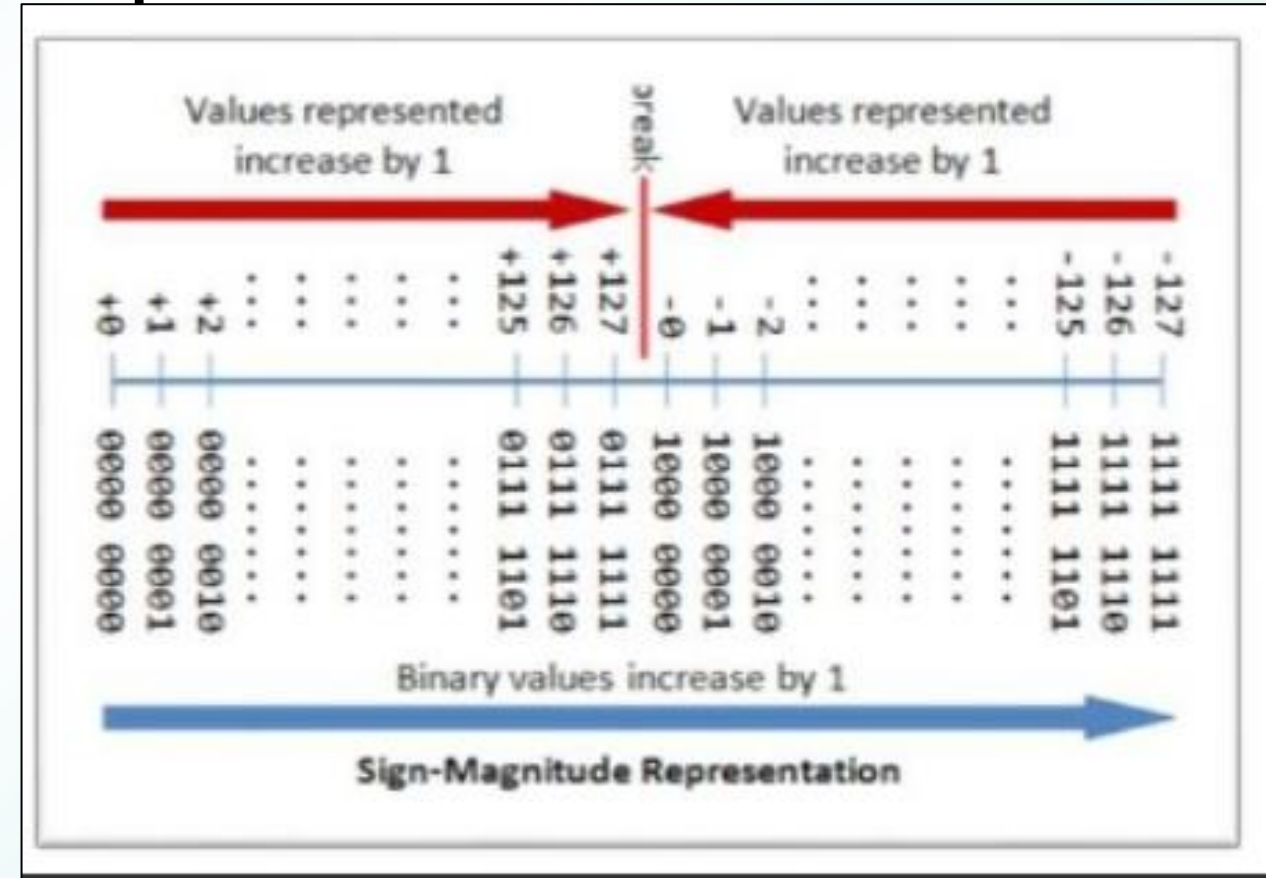
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Integer Representations and Operations

Sign and Magnitude Representation

- The MSB is the sign bit. 0 representing positive and 1 representing negative signs.
- The remaining (n-1) bits represent the magnitude (value) of the integer interpreted according to its binary pattern.
- Range is $-(2^{n-1}-1)$ to $+(2^{n-1}-1)$

Example:



Integer Representations and Operations

Sign and Magnitude Representation

Drawback:

- In one's complement representation, there are two different representations for zero:
 00000000 $\rightarrow$ +0
 11111111 $\rightarrow$ -0
- Since the processing method of positive and negative integers are different, the computer need to process the positive and negative numbers separately.



Integer Representations and Operations

Sign and Magnitude Representation

Activity 2

1. Using ONE byte, what is the maximum positive and negative integer that can be represented using sign and modulus?
2. Write the sign and modulus representation of the following integers:
 - 122
 - -100

Integer Representations and Operations

One's Complement Representation

- This representation of integers is used to simplify computer arithmetic.
- To represent positive and negative integers with absolute value less than 2^n , a total of $(n + 1)$ bits is used.
- The MSB is used to represent the sign. A '0' in this position is used for positive integers and a '1' in this position is used for negative integers.
- For positive integers, the remaining bits are identical to the binary expansion of the integer.
- For negative integers, the remaining bits are obtained by first finding the binary expansion of the absolute value of the integer, and then taking the complement of each of these bits, where the complement of a 1 is a 0 and the complement of a 0 is a 1.
- Range is $-(2^{n-1}-1)$ to $+(2^{n-1}-1)$



Integer Representations and Operations

One's Complement Representation

Example 1:

$$+100 = \mathbf{01100100}$$

$\rightarrow 100 < 2^7$, bits used 8

Example 2:

$$-126 = \mathbf{10000001}$$

a) $+126 = 01111110$

b) Swap bits $\rightarrow \mathbf{10000001} \rightarrow \mathbf{1\ 0000001}$

Sign bit

(n-1) bits

Integer Representations and Operations

One's Complement Representation

Drawback:

- In one's complement representation, there are two different representations for zero:

00000000 $\rightarrow +0$

11111111 $\rightarrow -0$

- Since the processing method of positive and negative integers are different, the computer need to process the positive and negative numbers separately.



Integer Representations and Operations

One's Complement Representation

Activity 3

Write the one's complement representation of the following integers:

- 0110111_2
- 00110011_2
- 22_{10}
- -125_{10}



Integer Representations and Operations

Two's Complement Representation

- This representation of integers is also used to simplify computer arithmetic (In fact, this representation is used much more commonly than one's complement representations).
- To represent an integer x with $-2^{n-1} \leq x \leq 2^{n-1} - 1$, n bits are used.
- The MSB is used to represent the sign. A '0' in this position is used for positive integers and a '1' in this position is used for negative integers.
- For positive integers, the remaining $(n-1)$ bits are identical to the binary expansion of the integer.
- For negative integers, the remaining $(n-1)$ bits are the bits of the binary expansion of $2^{n-1} - |x|$. Add leading zeroes, if necessary, to make the $(n-1)$ bits.
- Range is $-(2^{n-1})$ to $+(2^{n-1}-1)$



Integer Representations and Operations

Two's Complement Representation

Example 1: +100

8 bits are used $\rightarrow n = 8$

$$-2^{8-1} \leq 100 \leq 2^{8-1} - 1$$

Sign bit (n-1) bits

$\rightarrow +100 = 0 \ 1100100$

Example 2: -126

a) $2^{8-1} - |-126| = 2^7 - 126 = 127 - 126 = 1$

b) $1_{10} = 1 \rightarrow 0000001_2$

Sign bit (n-1) bits

1 0000010



Integer Representations and Operations

Two's Complement Representation

Advantages:

- There is only one representation for the number 0.
- In binary addition and subtraction, both positive and negative numbers can be treated together (processed together).
- Subtraction can use the addition logic.



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Integer Representations and Operations

Two's Complement Representation

Addition, Subtraction, Overflow and Underflow

Addition of positive integers

Example: $n=8$, $65D + 5D = 70D$

65D	→	0100 0001B
5D	→	0000 0101B
65D + 5D	→	0100 0110B OK 70D within the range of -128 to +127

Subtraction is the addition of positive and negative integers

Example: $n=8$, $65D - 5D = 60D \rightarrow 65D + (-5D)$

65D	→	0100 0001B
-5D	→	1111 1011B
65D + (-5D)	→	0011 1100B → 60D OK within the range of -128 to +127



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Integer Representations and Operations

Two's Complement Representation

Addition, Subtraction, Overflow and Underflow

Overflow and underflow happens when the result is not within the acceptable range

Example of Overflow: $n=8$, $127D + D = 129D$ (not within -128 and $+127$, exceed maximum value)

127D	→	0111 1111B
2D	→	0000 0010B
127D + 2D	→	1000 0001B → -127D wrong answer

Example of underflow: $n=8$, $-125D - 5D = -130D$ (not within -128 and $+127$, below the minimum value)

-125D	→	1000 0011B
-5D	→	1111 1011B
-125D + (-5D)	→	0111 1110B → +126D wrong answer



Integer Representations and Operations

Two's Complement Representation

- An n-bit two's complement integer can be represented from $-2^{(n-1)}$ to $+2^{(n-1)}-1$.
- This can represent all the integers within the range without missing any integers.

n	minimum	maximum
8	$-(2^7)$ ($=-128$)	$+(2^7)-1$ ($=+127$)
16	$-(2^{15})$ ($=-32,768$)	$+(2^{15})-1$ ($=+32,767$)
32	$-(2^{31})$ ($=-2,147,483,648$)	$+(2^{31})-1$ ($=+2,147,483,647$)(9+ digits)
64	$-(2^{63})$ ($=-9,223,372,036,854,775,808$)	$+(2^{63})-1$ ($=+9,223,372,036,854,775,807$)(18+ digits)

8bit from -128 to +127

16bit from -32,768 to +32,767

32bit from -2,147,483,648 to + 2,147,483,647

64bit from -9,223,372,036,854,775,808 to + 9,223,372,036,854,775,807



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Integer Representations and Operations

Two's Complement Representation

Decoding Two's Complement Numbers

- Check the sign bit first.
 - If sign bit = 0, the number is positive and the absolute value is the binary value of the remaining (n-1) integers.
 - If the sign bit = 1, the number is negative. To get the absolute value:
 - ❑ Invert remaining (n-1) bits and add 1 or
 - ❑ Scan from right until the first occurrence of 1. Flip all values from that point to the left. Binary value of this pattern gives the absolute value of the negative integer.

Example:

N=8 bit pattern = 1 100 0100 B

Negative number

Scan from right, until the first occurrence of 1, flip all values from that point to left

011 1100 B = 60 D

Represented negative value is -60 D



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Integer Representations and Operations

Two's Complement Representation

Activity 4

1. Write the two's complement representation of the following integers:

- 00101110_2
- 01101000_2
- -82_{10}

2. Decode the following two's complement number:

- 11110001_2



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Integer Representations and Operations

Operations on Integers

- Computers have a built-in limit on the size of integers that can be used in machine arithmetic.
- This upper limit is called the *word size*, denoted by w .
- The word size is usually a power of 2, such as 2^{32} or 2^{35} for Pentium machines.
- To store an integer $n > w$, express n in base w notation and for each digit of this expansion, one computer word is used.



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Integer Representations and Operations

Operations on Integers

- Binary Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ carry } 1$$

Examples:

$$\begin{array}{r} 1110_2 \\ + 0000 0110_2 \\ \hline 0001 0100_2 \end{array}$$

which is
which is

$$\begin{array}{r} 14_{10} \\ + 6_{10} \\ \hline 20_{10} \end{array}$$

$$14_{10} + (-6_{10}) = 8_{10}$$



$$14_{10} + (-6_{10}) = 8_{10} \rightarrow 14_{10} - (+6_{10}) = 8_{10}$$



Integer Representations and Operations

Operations on Integers

- Binary Subtraction

Perform just like normal subtraction of digits.
A borrow of 1 equals borrow of 2.

Examples:

$\begin{array}{r} 0000\ 1110_2 \\ - 0000\ 0110_2 \\ \hline 0000\ 1000_2 \end{array}$	which is which is	$\begin{array}{r} 14_{10} \\ - 6_{10} \\ \hline 8_{10} \end{array}$
--	----------------------	---

$\begin{array}{r} \overset{1}{0\ 10\ 10-10} \\ 000\cancel{1}\ \cancel{4}001_2 \\ - 0000\ 1110_2 \\ \hline 0000\ 1011_2 \end{array}$	$\rightarrow$ $\rightarrow$	$\begin{array}{r} 25_{10} \\ - 14_{10} \\ \hline 11_{10} \end{array}$
---	--------------------------------	---

- Working with numbers where the result is negative.

- The basic computational process is the same. Negative numbers are converted to their two's complement equivalent.
- But since the resulting number is negative, two's complement is used again to convert the number back to standard binary.



Integer Representations and Operations

Operations on Integers

- Binary Subtraction (Resulting to Negative Numbers)

Example:

$$9_{10} - 14_{10} = 9_{10} + (-14_{10}) = -5$$

Decimal	Standard Binary	
9_{10}	0000 1001 ₂	left-most digit says positive
$+ (-14_{10})$	$+ 1000 1110_2$	left-most digit says negative
$- 5_{10}$		

Step 1: Decide on the number of binary digits -- Choose 8

Step 2: Convert the negative number to its two's complement equivalent.

Write down the negative number with sign bit set to 0	0000 1110 ₂
A) Flip the digits	1111 0001 ₂
B) Add one (1)	$+ 1$
C) This is two's complement	1111 0010 ₂

Step 3: Add the two numbers together.

9_{10}	0000 1001 ₂	
$+ (-14_{10})$	$+ 1111 0010_2$	in two's complement format
	1111 1011 ₂	no overflow, this says result is still in two's complement form

Step 4: Convert from two's complement by doing the two's complement process again.

Write down the number	1111 1011 ₂	
A) Flip the digits	0000 0100 ₂	
B) Add one (1)	$+ 1$	
C) This is two's complement	0000 0101 ₂	positive 5, in standard binary
D) We know the result Is negative, change sign bit	1000 0101 ₂	negative 5, in standard binary



Integer Representations and Operations

Operations on Integers

- Binary Multiplication

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Example:

To multiply $(1101)_2$ and $(1110)_2$

$$\begin{array}{r} 1101 \\ \times 1110 \\ \hline 0000 \\ 1101 \\ 1101 \\ 1101 \\ \hline 10110110 \end{array}$$



Integer Representations and Operations

Operations on Integers

- Binary Division

$$1 \div 1 = 1$$

$$1 \div 0 = 0$$

$$0 \div 1 = \text{Meaningless}$$

$$0 \div 0 = \text{Meaningless}$$

Four-step Process:

- Divide
- Multiply
- Subtract
- Bring down

Example:

Handwritten long division of 128 by 8. The divisor 8 is written on the left. The dividend 128 is written under the division bar. The quotient 016 is written above the bar. The steps show: 8 goes into 12 one time (01), 8 goes into 8 one time (016), and the remainder is 0.

Handwritten long division of 1101010 by 110. The divisor 110 is written on the left. The dividend 1101010 is written under the division bar. The quotient 0001111 is written above the bar. The steps show: 110 goes into 1101 one time (0001), 110 goes into 1010 one time (00011), 110 goes into 1010 one time (000111), and the remainder is 0.



Integer Representations and Operations

Operations on Integers

Activity 5

Perform the following binary arithmetic:

1) $101111011_2 + 1100111011_2$

2) $9_{10} - 7_{10} = 2_{10}$

3) Multiply 11101_2 and 110001_2

4) Find the quotient and remainder when $(110011111)_2$ is divided by $(1101)_2$



Formative Assessment 5

Check Canvas



SUBMODULE 2

Divisibility Theory in the Integers



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OBJECTIVES

- Demonstrate understanding of the concepts of divisibility.
- Differentiate prime numbers from composite numbers.
- Apply divisibility properties and the Fundamental Theorem of Arithmetic.



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Divisibility Theory in the Integers

Divisibility

Let n and d be integers, $d \neq 0$.

It is said that **d divides n** (written as $d \mid n$) if there exists an integer **q** satisfying **$n = dq$** .

where: q is called the *quotient*
 d is called a *divisor* or a *factor* of n .

Note:

- If d does not divide n , it is written as $d \nmid n$.
- When a divides b we say that a is a factor of b and that b is a multiple of a .



Divisibility Theory in the Integers

Divisibility Theorems

Let m , n and d be integers

(a) If $d \mid m$ and $d \mid n$ then $d \mid (m + n)$

Example: $3 \mid 9$ and $3 \mid 6$ then $3 \mid (9 + 6) \rightarrow 3 \mid 15$

(b) If $d \mid m$ and $d \mid n$ then $d \mid (m - n)$

Example: $3 \mid 9$ and $3 \mid 6$ then $3 \mid (9 - 6) \rightarrow 3 \mid 3$

(c) If $d \mid m$ then $d \mid mn$

Example: $5 \mid 10$ then $5 \mid (10)(2) \rightarrow 5 \mid 20$



Divisibility Theory in the Integers

Division Algorithm

Theorem:

For any integers **a** and **b** with $a \neq 0$ there exist unique integers **q** and **r** such that

$$\mathbf{b} = \mathbf{a}q + \mathbf{r}, \text{ where } 0 \leq r < |a|.$$

In the above equation, q is called the *quotient*, and r is called the *remainder*.

Examples:

Let $b = 49$ and $a = 4 \rightarrow 49 = (4)(12) + 1 \rightarrow$ Quotient is 12 and remainder is 1

Let $b = 17$ and $a = 5 \rightarrow 17 = (5)(3) + 2 \rightarrow$ Quotient is 3 and remainder is 2

Let $b = -11$ and $a = 3 \rightarrow -11 = (3)(-4) + 1 \rightarrow$ Quotient is -4 and remainder is 1



Divisibility Theory in the Integers

Prime and Composite Numbers

Prime Number

- An integer greater than 1 whose only positive divisors are itself and 1.

Examples: 2, 3, 5, 7

Composite Number

- An integer greater than 1 that is not prime.

Examples: 4, 6, 8, 9

Divisibility Theory in the Integers

Prime and Composite Numbers

Algorithm to test whether an integer is prime

```
input n
output d

is_prime(n)
{
    for d = 2 to  $\lfloor \sqrt{n} \rfloor$ 
        if ( n mod d == 0 )
            return d
    return 0
}
```

The algorithm determines whether the integer $n > 1$ is prime. If n is prime, the algorithm return 0. If n is composite, the algorithm returns a divisor d satisfying $2 \leq d \leq \sqrt{n}$.

$\lfloor x \rfloor$ means floor of x (the greatest integer less than or equal to x)

Examples:

a) 43

$\sqrt{43} = 6.55748$
 $\lfloor \sqrt{43} \rfloor = 6$
Test $d = 2$ to 6

b) 451

$\sqrt{451} = 21.23676$
 $\lfloor \sqrt{451} \rfloor = 21$
Test $d = 2$ to 21



Divisibility Theory in the Integers

Prime and Composite Numbers

Activity 6

1. Show that $3 \mid 99$.
2. Using division algorithm, show that 7 does not divide 1717.
3. Determine whether 121 is a prime number using the algorithm for testing whether an integer is prime or not.



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Divisibility Theory in the Integers

Prime and Composite Numbers

Relatively Prime Integers

- Two integers are relatively prime if their greatest common divisor is 1.

Examples: 15 and 28 are relatively prime
35 and 28 not relatively prime

Pairwise Relatively Prime Integers

- A list of integers are pairwise relatively prime if every pair of the items in the list are relatively prime.

Examples: 15, 17 and 28 are pairwise relatively prime
15, 17 and 27 are not pairwise relatively prime



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Divisibility Theory in the Integers

Primes and Their Distributions

Fundamental Theorem of Arithmetic

- Every positive integer can be written **uniquely** as the **product of primes**, where the prime factors are written in order of increasing size.

$$15 = 3 \cdot 5$$

$$48 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 = 2^4 \cdot 3$$

$$17 = 17$$

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$$

$$512 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^9$$

$$515 = 5 \cdot 103$$

$$28 = 2 \cdot 2 \cdot 7$$

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Divisibility Theory in the Integers

Primes and Their Distributions

Fermat's Little Theorem

- Given in 1640 by French mathematician **Pierre de Fermat**.

Let p be a prime number and n is any whole number then $p \mid n^p - n$ for any integer $n \geq 1$.

- A Fermat prime is a prime number of the form $2^{2^n} + 1$ for some nonnegative integer n .
- The first five Fermat numbers (for $n = 0, 1, 2, 3, 4$) are:

$$F_0 = 2^{2^0} + 1 = 3$$

$$F_1 = 2^{2^1} + 1 = 5$$

$$F_2 = 2^{2^2} + 1 = 17$$

$$F_3 = 2^{2^3} + 1 = 257$$

$$F_4 = 2^{2^4} + 1 = 65537$$



Divisibility Theory in the Integers

Primes and Their Distributions

Mersenne Prime

- The numbers are named for the French theologian and mathematician **Marin Mersenne**.
- A Mersenne prime is a prime number of the form $2^n - 1$ where n is a natural number.
- At present only 43 Mersenne prime numbers are known.
- Some of which are as follows:

2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, 607



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Divisibility Theory in the Integers

Primes and Their Distributions

Sieve of Erastosthenes

- A simple and ancient algorithm used to find the prime numbers up to any given limit. It is one of the most efficient ways to find small prime numbers.
- For a given upper limit n the algorithm works by iteratively marking the multiples of primes as composite, starting from 2. Once all multiples of 2 have been marked composite, the multiples of next prime, i.e. 3 are marked composite. This process continues until $p \leq \sqrt{n}$ where p is a prime number.



Divisibility Theory in the Integers

Primes and Their Distributions

Sieve of Erastosthenes Algorithm

In the following algorithm, the number 0 represents a composite number.

1. To find out all primes under n , generate a list of all integers from 2 to n . (Note: 1 is not prime)
2. Start with a smallest prime number, ie $p = 2$.
3. Mark all the multiples of p which are less than n as composite. To do this, mark the value of the numbers (multiples of p) in the generated list as 0. Do not mark p itself as composite.
4. Assign the value of p to the next prime. The next prime is the next non-zero number in the list which is greater than p .
5. Repeat the process until $p \leq \sqrt{n}$.



Divisibility Theory in the Integers

Primes and Their Distributions

Sieve of Eratosthenes Example

Generate all the primes less than 11.

1. Create a list of integers from 2 to 10. `list = [2, 3, 4, 5, 6, 7, 8, 9, 10]`
2. Start with $p = 2$.
3. Since $2^2 \leq 10$, continue.
4. Mark all multiples of 2 as composite by setting their value as 0 in the list. `list = [2, 3, 0, 5, 0, 7, 0, 9, 0]`
5. Assign the value of p to the next prime, ie 3.
6. Since $3^2 \leq 10$, continue.
7. Mark all multiples of 3 as composite by setting their value as 0 in the list. `list = [2, 3, 0, 5, 0, 7, 0, 0, 0]`
8. Assign the value of p to 5.
9. Since $5^2 \not\leq 10$, stop.

Prime numbers are 2, 3, 5, 7



Divisibility Theory in the Integers

Primes and Their Distributions

Activity 7

1. Are 67942 and 4209 relatively prime?
2. Are 121, 122, and 123 pairwise relatively prime?
3. Using Sieve of Erastosthenes algorithm, generate all primes less than 25.



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SUBMODULE 3

Prime Factorization, GCD and LCM



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OBJECTIVES

- Define prime factorization
- Identify methods for prime factorization
- Define GCD and LCM
- Identify methods for solving GCD and LCM



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Prime Factorization, GCD and LCM

“Prime factorization” is finding which prime numbers multiply together to make the original number.

Example 1:

What are the prime factors of 12?

It is best to start working from the smallest prime number, which is 2:

$$12 / 2 = 6$$

But 6 is not a prime number, so there is a need to go further.

$$6 / 2 = 3$$

Since 3 is a prime number, answer is obtained.

Thus, prime factorization of 12 is $2 \times 2 \times 3$ or **$12 = 2^2 \times 3$**

Prime Factorization, GCD and LCM

Prime Factorization

Example 2: What is the prime factorization of 147 ?

Can we divide 147 exactly by 2?

$$147 \div 2 = 73\frac{1}{2}$$

No it can't. The answer should be a whole number, and $73\frac{1}{2}$ is not.

Let's try the next prime number, 3:

$$147 \div 3 = 49$$

That worked, now we try factoring 49.

The next prime, 5, does not work. But 7 does, so we get:

$$49 \div 7 = 7$$

And that is as far as we need to go, because all the factors are prime numbers.

$$\mathbf{147 = 3 \times 7 \times 7}$$

(or $\mathbf{147 = 3 \times 7^2}$ using exponents)



Prime Factorization, GCD and LCM

Prime Factorization

Example 3: What are the prime factors of 90 ?

Break 90 into 9×10

- The prime factors of 9 are **3 and 3**
- The prime factors of 10 are **2 and 5**

So the prime factors of 90 are **3, 3, 2 and 5**



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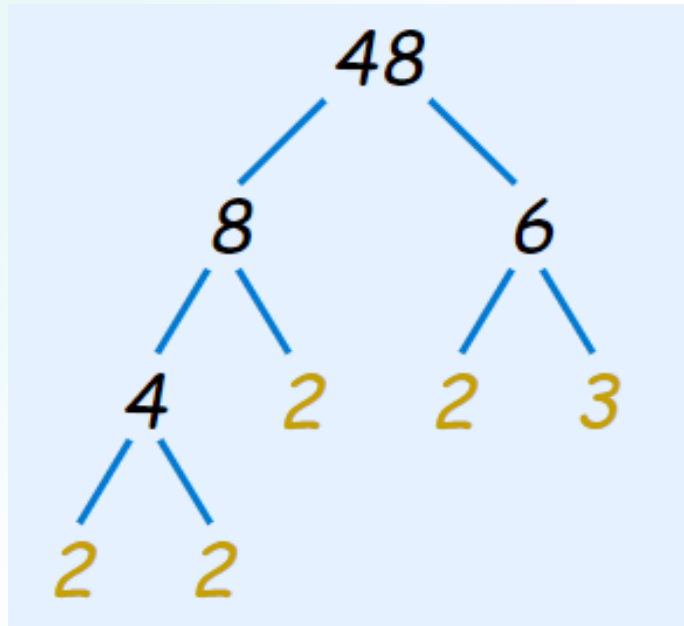
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Prime Factorization, GCD and LCM

A **factor tree** shows the prime factors of a composite number in a "tree-like" form. Drawing factor trees is a good way of doing prime factorization.

Example : Factor tree of 48.



$$48 = 2 \times 2 \times 2 \times 2 \times 3$$

(or $48 = 2^4 \times 3$ using exponents)



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Prime Factorization, GCD and LCM

Activity 8

1. Determine the prime factors of 56 using prime factorization.
2. Determine the prime factors of 96 using a factor tree.



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Prime Factorization, GCD and LCM

The **greatest common divisor (gcd)** of two or more integers, which are not all zero, is the largest positive integer that divides each of the integers.

Various notational conventions are summarized in the following table:

Notation	Source
$\text{GCD}(a, b)$	Zwillinger, D. (Ed.). "Greatest Common Divisor." §2.3.5 in <i>CRC Standard Mathematical Tables and Formulae</i> , 30th ed. Boca Raton, FL: CRC Press, p. 91, 1996.
$\text{gcd}(a, b)$	Gellert, W.; Gottwald, S.; Hellwich, M.; Kästner, H.; and Küstner, H. (Eds.). <i>VNR Concise Encyclopedia of Mathematics</i> , 2nd ed. New York: Van Nostrand Reinhold, p. 25, 1989.
$\text{g.c.d. } (a, b)$	Andrews, G. E. <i>Number Theory</i> . New York: Dover, p. 22, 1994.
(a, b)	



Prime Factorization, GCD and LCM

The Greatest Common Divisor

1. $\text{GCD}(48, 72)$

The positive common divisors are 1, 2, 3, 4, 6, 8, 12, 16 and 24.
Therefore $\text{GCD}(48, 72) = 24$.

2. $\text{GCD}(19, 72)$

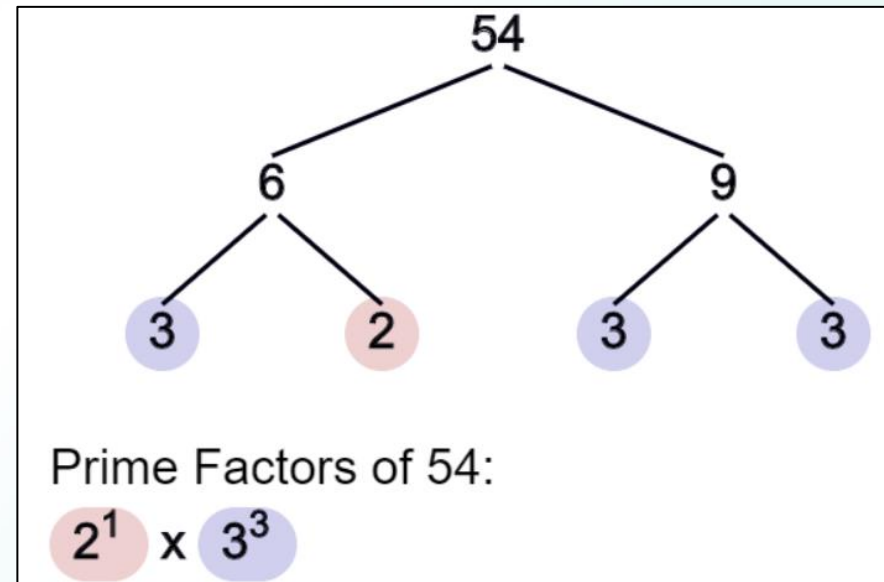
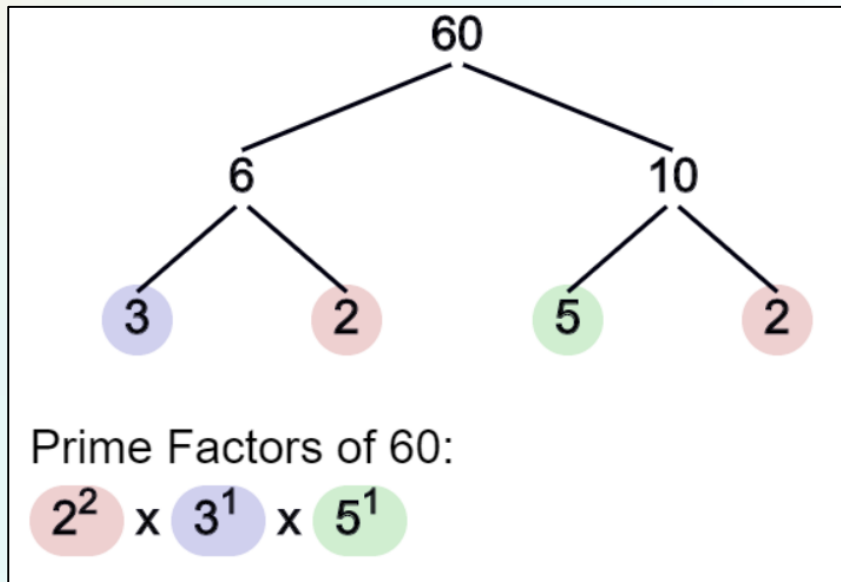
The only positive common divisor is 1.
Therefore $\text{GCD}(19, 72) = 1$.

Prime Factorization, GCD and LCM

GCD Using Factor Trees

- The GCD is the product of the common prime factors.

Example 1: GCD(60, 54)



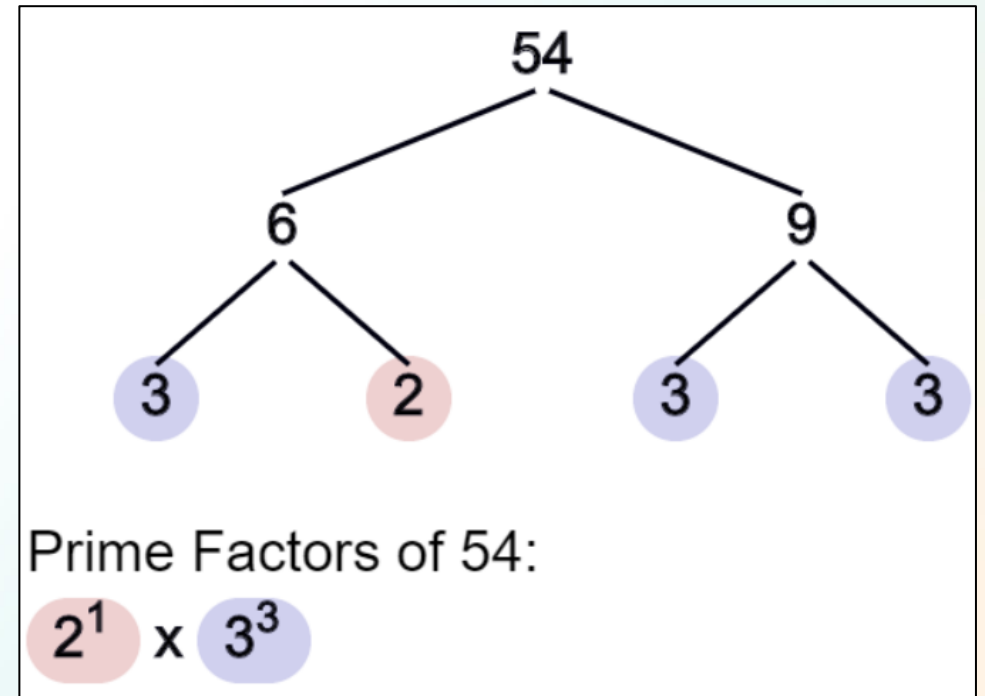
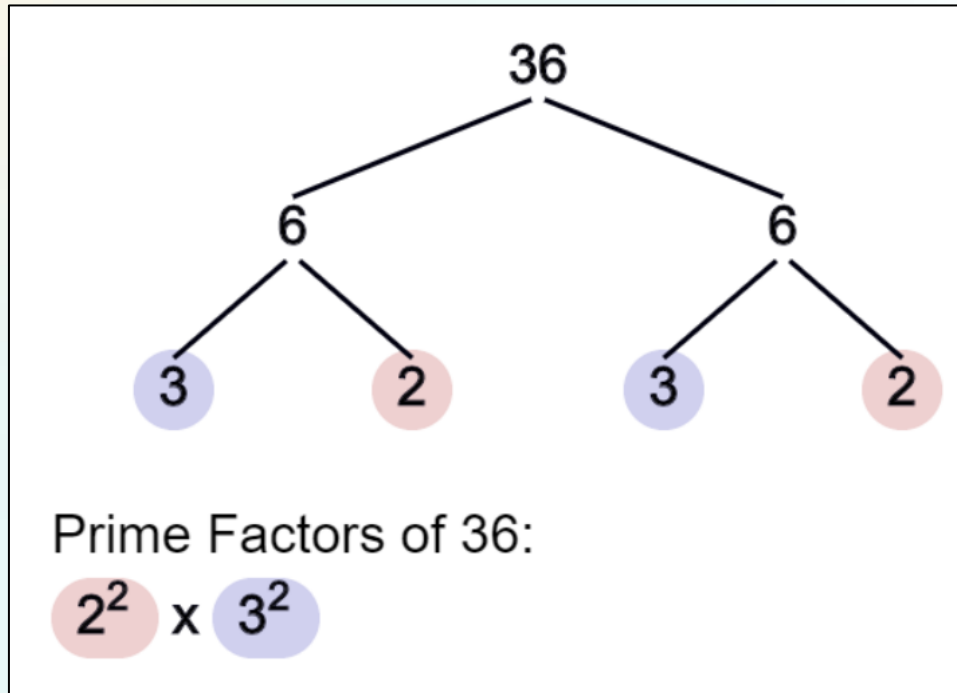
Common factor: 3^1 and 2^1 . $\text{GCD}(60, 54) = 2 \times 3 = 6$



Prime Factorization, GCD and LCM

GCD Using Factor Trees

Example 2: $\text{GCD}(36, 54)$



Common factor: 3^2 and 2^1 . $\text{GCD}(36, 54) = 2^1 \times 3^2 = \mathbf{18}$

Prime Factorization, GCD and LCM

GCD Using Euclidean Algorithm

The **Euclidean Algorithm** is a technique for quickly finding the **GCD** of two integers.

Lemma:

If **a , b , q , r** are integers and **$a = bq + r$** , then $GCD(a, b) = GCD(b, r)$.

The idea is to keep repeating the division algorithm.

Prime Factorization, GCD and LCM

GCD Using Euclidean Algorithm

Example 1: GCD(252, 198)

Solution: By the Euclidean Algorithm we have

$$252 = 198 \cdot 1 + 54$$

$$198 = 54 \cdot 3 + 36$$

$$54 = 36 \cdot 1 + 18$$

$$36 = 18 \cdot 2$$

therefore $(252, 198) = 18$.



GCD Using Euclidean Algorithm

Example 2: GCD(326, 78)

Solution: By the Euclidean Algorithm we have

$$326 = 78 \cdot 4 + 14$$

$$78 = 14 \cdot 5 + 8$$

$$14 = 8 \cdot 1 + 6$$

$$8 = 6 \cdot 1 + 2$$

$$6 = 2 \cdot 3$$

therefore $(326, 78) = 2$.



Prime Factorization, GCD and LCM

Linear Diophantine Equation

A **Diophantine equation** is a polynomial equation whose solutions are restricted to integers.

The simplest type of a Diophantine equation is the **linear Diophantine equation** in two unknowns, which takes the form:

$$ax + by = c$$

where a, b, c are given integers and a, b are not both zero.

A solution of this equation is a pair of integers x_0, y_0 that, when substituted into the equation, satisfy it. That is $ax_0 + by_0 = c$.

Prime Factorization, GCD and LCM

Linear Diophantine Equation

Theorem:

The linear Diophantine equation $ax + by = c$ has a solution if and only if $d \mid c$, where $d = \text{GCD}(a, b)$.

Example: $172x + 20y = 1000$

Applying the Euclidean' Algorithm to get $\text{GCD}(172, 20)$:

$$172 = 8 \cdot 20 + 12$$

$$20 = 1 \cdot 12 + 8$$

$$12 = 1 \cdot 8 + 4$$

$$8 = 2 \cdot 4$$

$$\text{GCD}(172, 20) = 4$$

Since $4 \mid 1000$, therefore a solution exists.



Prime Factorization, GCD and LCM

Linear Diophantine Equation

To obtain the integer 4 as a linear combination of 172 and 20, move backwards through the previous calculations:

$$172 = (8)(20) + 12 \quad \rightarrow (1)(172) - (8)(20) = 12 \quad \text{eq. 1}$$

$$20 = (1)(12) + 8 \quad \rightarrow (1)(20) - (1)(12) = 8 \quad \text{eq. 2}$$

$$12 = (1)(8) + 4 \quad \rightarrow (1)(12) - (1)(8) = 4 \quad \text{eq. 3}$$

Working backwards:

$$(1)(12) - (1)(8) = 4$$

from eq. 3

$$(1)(12) - (1) [(1)(20) - (1)(12)] = 4$$

from eq. 2

$$(1)(12) - (1)(20) + (1)(12) = 4$$

$$(2)(12) - (1)(20) = 4$$

$$(2) [(1)(172) - (8)(20)] - (1)(20) = 4 \quad \text{from eq. 1}$$

$$(2)(172) - (16)(20) - (1)(20) = 4$$

$$(2)(172) - (17)(20) = 4$$

$$[(2)(172) - (17)(20) = 4] \times 250$$

$$(172)(500) - (20)(4250) = 1000$$

$$(172)(500) + (20)(-4250) = 1000$$

$$x = 500, y = -4250$$



Prime Factorization, GCD and LCM

Activity 9

1. Determine $\text{GCD}(315, 825)$ using Factor Tree.
2. Determine $\text{GCD}(4361, 42371)$ using Euclidean Algorithm.
3. Find x and y in the linear Diophantine equation $65x + 40y = 5$.

Prime Factorization, GCD and LCM

The **least common multiple (lcm)** of two or more integers, which are not all zero, is the smallest positive integer that divides each of the integers.

Example: LCM(4, 6)

Multiples of 4 are:

4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, ...

Multiples of 6 are:

6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, ...

Common multiples of 4 and 6 are the numbers that are in both lists:

12, 24, 36, 48, 60, 72,

So, from this list of the first few common multiples of the numbers 4 and 6, their *least common multiple* is 12.



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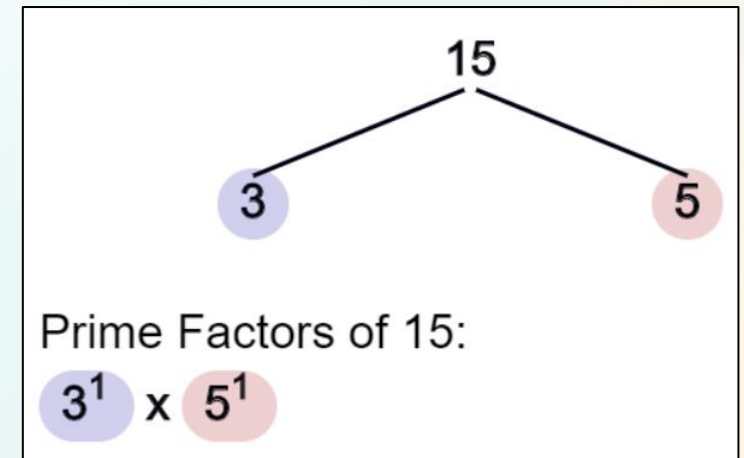
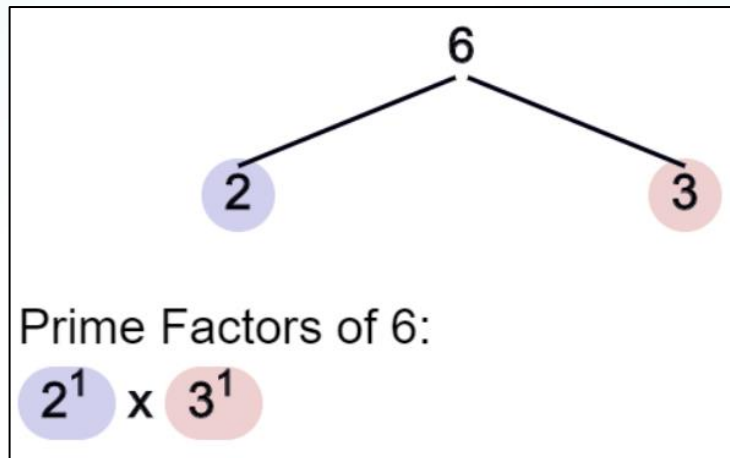
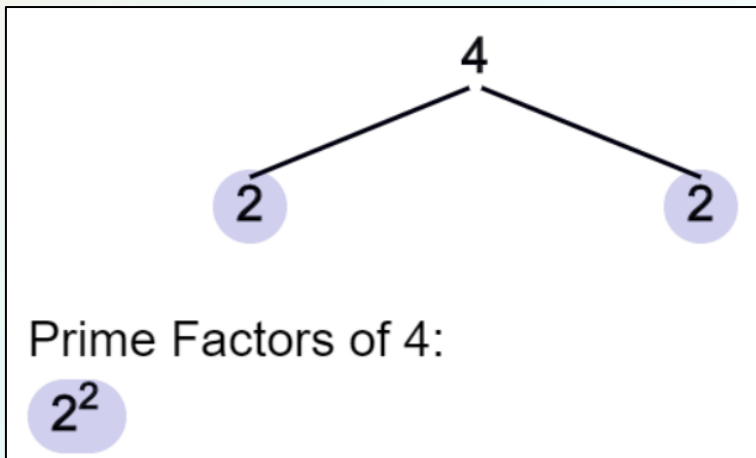
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Prime Factorization, GCD and LCM

LCM Using Factor Tree

- The LCM is the product of the largest multiple of each prime that appears on at least one list.

Example : LCM(4, 6, 15)



$$\text{LCM}(4, 6, 15) = 2^2 \times 3^1 \times 5^1 = \mathbf{60}$$

Prime Factorization, GCD and LCM

LCM Using Ladder (Cake) Method

Example: Find the LCM(10, 12, 15, 75)

1. Write down the numbers in a cake layer (row).

Cake / Ladder

10	12	15	75
----	----	----	----

2. Divide the layer numbers by a prime number that is evenly divisible into two or more numbers in the layer and bring down the result into the next layer.

Cake / Ladder

2	10	12	15	75
	5	6		

3. If any number in the layer is not evenly divisible just bring down that number.

Cake / Ladder

2	10	12	15	75
	5	6	15	75

4. Continue dividing cake layers by prime numbers.
5. When there are no more primes that evenly divided into two or more numbers then its done.

Cake / Ladder

2	10	12	15	75
3	5	6	15	75
5	5	2	5	25
	1	2	1	5

- The LCM is the product of the numbers in the L shape, left column and bottom row. 1 is ignored.
- $LCM = 2 \times 3 \times 5 \times 2 \times 5$
- $LCM = 300$
- Therefore, $LCM(10, 12, 15, 75) = 300$



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Prime Factorization, GCD and LCM

GCD and LCM Theorems

Theorem:

$$a * b = \text{GCD}(a, b) * \text{LCM}(a, b)$$

Example:

$$a = 60$$

$$b = 54$$

$$\text{GCD}(60, 54) = 6$$

$$\text{LCM}(60, 54) = 540$$

$$(60)(54) = (6)(540)$$

$$3240 = 3240$$



Prime Factorization, GCD and LCM

Least Common Multiple (LCM)

Activity 10

1. Determine $\text{LCM}(24, 32, 40)$ using Factor Tree.
2. Determine $\text{LCM}(175, 60, 128)$ using Ladder Method.



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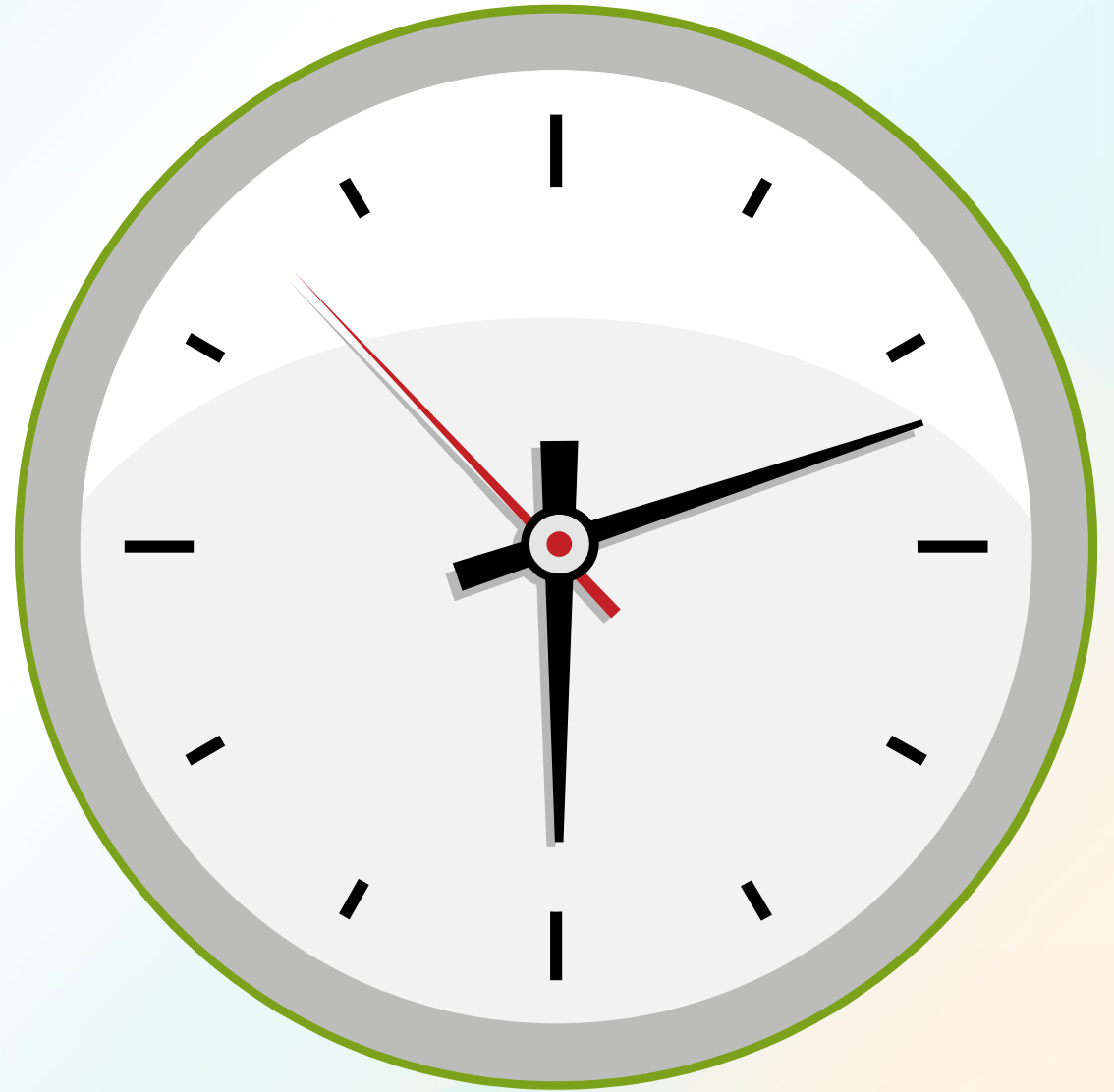
Formative Assessment 7

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QUIZ REVIEW

SUMMATIVE ASSESSMENT 2



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